

DATABASE SYSTEMS • COURSE 10127

Zoo Management System

From Java Objects to a PostgreSQL Database

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Individual project (approved)

What Is This Project?

BEFORE

Java OOP Program

- An object-oriented zoo built in a previous course
- Animals modeled with classes & inheritance
- Data kept in arrays and files in memory
- Lost on every restart — no real storage

AFTER

PostgreSQL Database

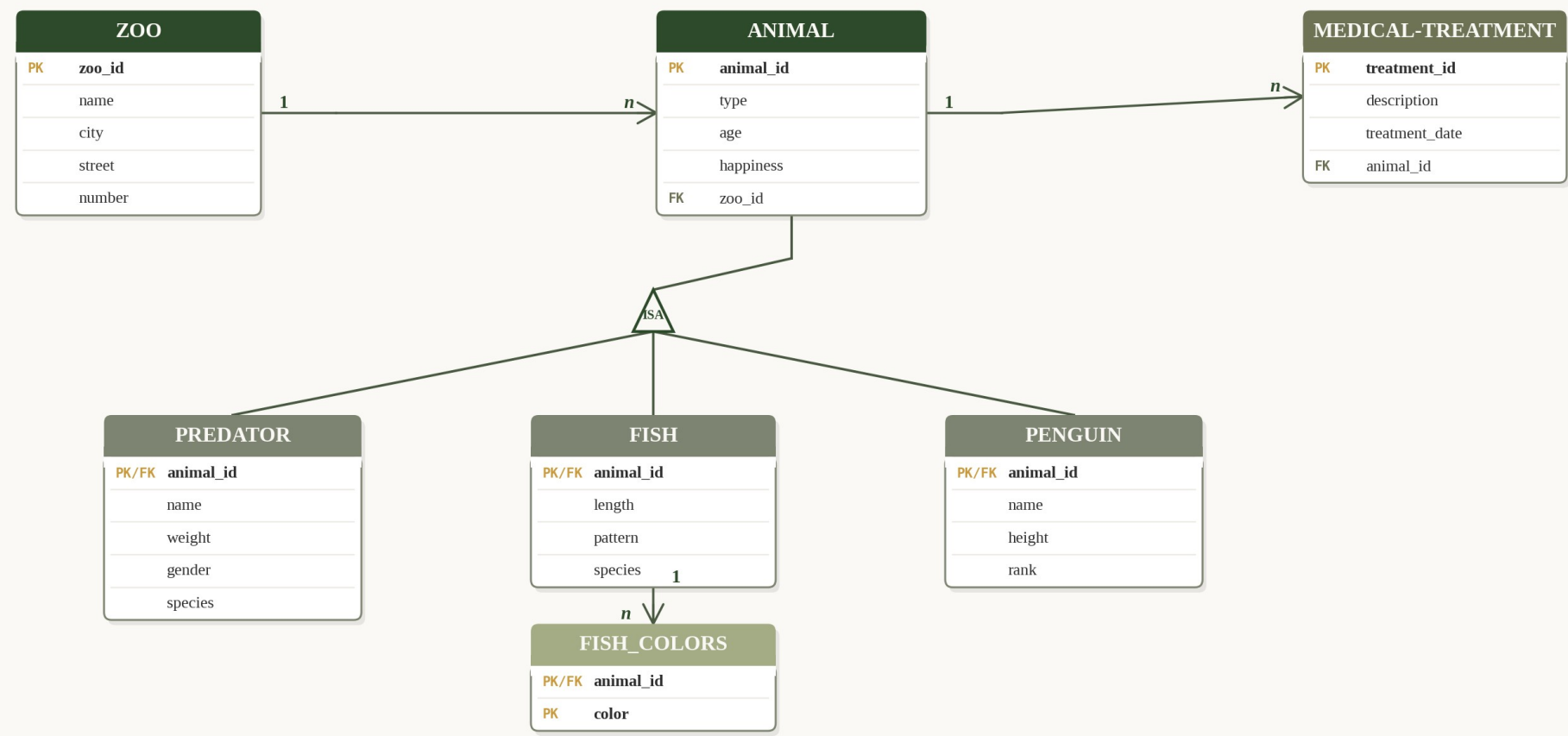
- A normalized relational schema (7 tables, 3NF)
- All animal data stored permanently in Postgres
- Java connects via JDBC and runs full CRUD
- Data survives restarts and supports SQL queries

The Goal

Take an existing OOP program and give it a real database back end. The focus is the **ERD**, the **tables**, and the **interaction** between the Java program and PostgreSQL through JDBC.

Entity-Relationship Diagram

ISA hierarchy: every animal has one row in ANIMAL plus one row in its specific type table (shared animal_id).



The Tables (7 total · 3NF)

Table	Purpose	Key design point
ZOO	The zoo itself (name, address)	Parent of all animals
ANIMAL	Base table for every animal	Holds shared fields: age, happiness
PREDATOR	Lions & tigers	Merged via species column (no duplicate tables)
FISH	All fish types	species column instead of subclasses
FISH_COLORS	Colors of each fish	Composite key resolves Color[] → 1NF
PENGUIN	Penguins	rank replaces the Java linked-list order
MEDICALTREATMENT	Vet treatments	One FK to ANIMAL serves every animal type

Why merge subclasses? OOP inheritance does not map 1-to-1 to tables. Lion vs Tiger had no unique columns — only different behaviour — so they became one `species` value instead of separate tables.

Beyond the Requirements

1

JDBC Integration

- PreparedStatement + ResultSet
- Full CRUD on every table
- Runtime password via Scanner
- Protected against SQL injection

2

10+ SQL Queries

- JOINS across ANIMAL & subtypes
- Aggregates: COUNT, AVG happiness
- GROUP BY species
- Sub-queries for vet treatments

3

Trigger (Bonus)

- AFTER INSERT on MEDICALTREATMENT
- Auto-updates animal state
- Logic lives in the database
- Shows transaction awareness

LIVE DEMO Run the Java program → connect to PostgreSQL → insert an animal → see it appear with a SQL query in DataGrip.

The SQL Queries (10 total)

Written and tested directly in DataGrip, then run through JDBC via ZooQueries.java

#	What It Answers	SQL Technique	Query (condensed)
1	Predators + happiness level	JOIN	<code>FROM Animals a JOIN PREDATOR p ON p.AnimalID=a.AnimalID</code>
2	Each fish + its colors	JOIN	<code>FROM FISH f JOIN FISH_COLORS fc ON fc.AnimalID=f.AnimalID</code>
3	Animals + medical treatments	JOIN	<code>FROM Animals a JOIN MEDICAL_TREATMENT mt ON mt.AnimalID=a.AnimalID</code>
4	Animal count by type	GROUP BY + COUNT	<code>GROUP BY AnimalType -- COUNT(*)</code>
5	Average age by type	GROUP BY + AVG	<code>GROUP BY AnimalType -- AVG(Age)</code>
6	Number of colors per fish	JOIN + GROUP BY + COUNT	<code>GROUP BY f.AnimalID -- COUNT(fc.FishColor)</code>
7	Happiest animals (Happiness > 90)	WHERE + ORDER BY	<code>WHERE Happiness > 90 ORDER BY Happiness DESC</code>
8	The heaviest predator	ORDER BY + LIMIT	<code>ORDER BY PredatorWeight DESC LIMIT 1</code>
9	Penguins ordered by rank	ORDER BY	<code>ORDER BY PenguinRank</code>
10	Animal type with highest avg age	GROUP BY + ORDER BY + LIMIT	<code>GROUP BY AnimalType ORDER BY AVG(Age) DESC LIMIT 1</code>

Use of AI in This Project

I used Claude (AI) to help me understand things and guide me — not to write the project for me.

Task	Prompt	Was it useful?	Did I need to change it?
ERD design	Asked why Lion/Tiger shouldn't be separate tables	Yes — helped me see why	Yes — changed my first design
Choosing SQL queries	Asked what queries make sense for this database	Yes — talked through it	Small changes after testing
CREATE TABLE + 3NF	Asked for help understanding normalization and writing the SQL	Yes — explained it, guided the SQL	Some, based on my schema
JDBC connection issues	Asked why my program couldn't connect to PostgreSQL	Yes — helped me find the real cause	No — fixed it myself
Trigger (bonus)	Asked what a trigger is and how to use one here	Yes — explained it clearly	No — worked once tested
Presentation	Asked for help building slides showing the process	Yes — helped build the slides	Some formatting changes

How I tested it: I ran everything myself in DataGrip to check it worked the way I expected. For the trigger, I inserted a real medical treatment and checked the animal's happiness number went up correctly. For the JDBC problems, I ran the program again after every fix to make sure it actually connected.